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C6 — Global Challenges

OCR GCSE Chemistry A (J248) — Gateway Science · Papers 2 & 4

Topic C6 is **synoptic**: it applies ideas from C1–C5 (reactivity, electrolysis, rates, equilibrium, bonding) to real-world problems — making metals and fertilisers, choosing materials, organic chemistry and the Earth's atmosphere and water supply.

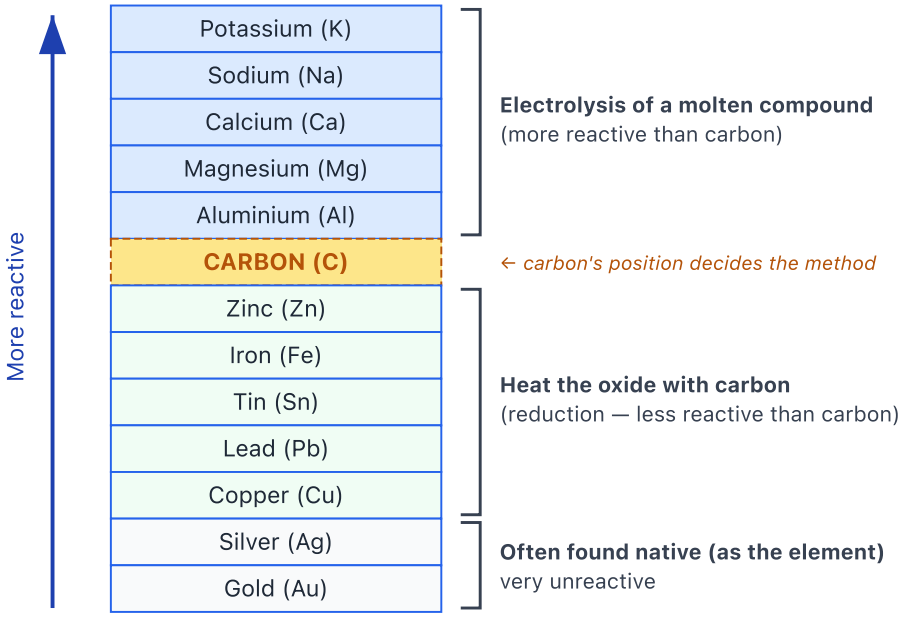
C6.1 Improving Processes and Products

Extracting Metals Using Carbon — C6.1a

> Most metals are found in the Earth's crust as compounds in **ores** (rocks containing enough metal compound to make extraction worthwhile). Extraction is a **reduction** — the metal compound loses oxygen. The method depends on the metal's position in the **reactivity series relative to carbon**.

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Reactivity Series and Method of Extraction



- **Iron** (a ferrous metal) is extracted in a blast furnace. Coke (carbon) burns to make carbon monoxide, which reduces iron(III) oxide:
  $C + O_2 \rightarrow CO_2$ · $CO_2 + C \rightarrow 2CO$ · **$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$**
- **Non-ferrous metals** below carbon are extracted the same way, e.g. copper and zinc:
  $2CuO + C \rightarrow 2Cu + CO_2$ $ZnO + C \rightarrow Zn + CO$
- Carbon is used because it is **cheap**, plentiful and more reactive than these metals, so it takes the oxygen from them. The carbon is **oxidised**; the metal oxide is **reduced**.

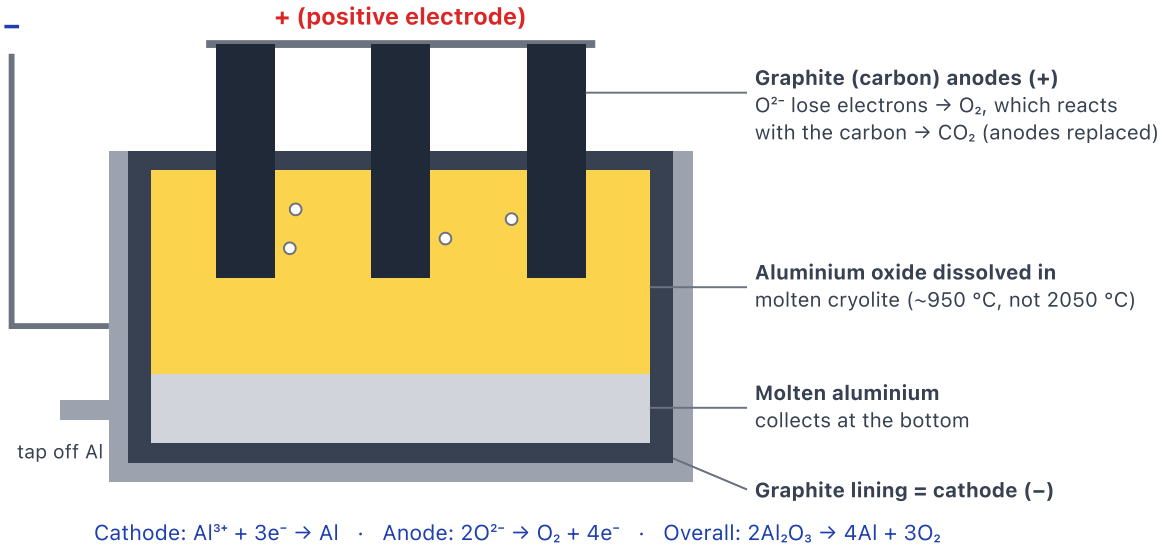
> **Worked Example:** PAG C1: heat a mixture of black copper(II) oxide and carbon powder strongly in a crucible or ignition tube. Brown/pink copper forms, showing that carbon is more reactive than copper. Tip the cooled product into water to separate the dense copper from unreacted carbon.

Extracting Metals by Electrolysis — C6.1b

Metals **more reactive than carbon** (e.g. aluminium, sodium, magnesium) cannot be reduced by carbon, so they are extracted by **electrolysis** of the molten compound. This uses a lot of electrical energy, so it is expensive.

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Extraction of Aluminium by Electrolysis



Cathode: $Al^{3+} + 3e^- \rightarrow Al$ · Anode: $2O^{2-} \rightarrow O_2 + 4e^-$ · Overall: $2Al_2O_3 \rightarrow 4Al + 3O_2$

- Aluminium oxide (from the ore **bauxite**) has a very high melting point (about 2050 °C). Dissolving it in molten **cryolite** lowers the operating temperature to about 950 °C, saving energy. The ions must be free to move, so the compound must be molten or dissolved.
- Positive Al^{3+} ions move to the negative cathode and gain electrons; negative O^{2-} ions move to the positive anode and lose electrons.
- The oxygen reacts with the carbon anodes to form carbon dioxide, so the anodes must be **replaced regularly**.
- Electrolysis of **aqueous** solutions (PAG C2: aqueous sodium chloride or copper(II) sulfate, testing the gases formed) is also used industrially, e.g. to purify copper and to make chlorine and sodium hydroxide.

> **Exam Tip:** Extraction by electrolysis costs more than reduction with carbon because of the large amount of **electricity** needed to melt the compound and to decompose it. That is why aluminium is worth recycling.

Biological Methods of Metal Extraction (HT) — C6.1c

High-grade copper ores are running out. **Low-grade ores** (and waste from old mines) contain too little copper to extract economically by traditional methods. Two biological methods can be used:

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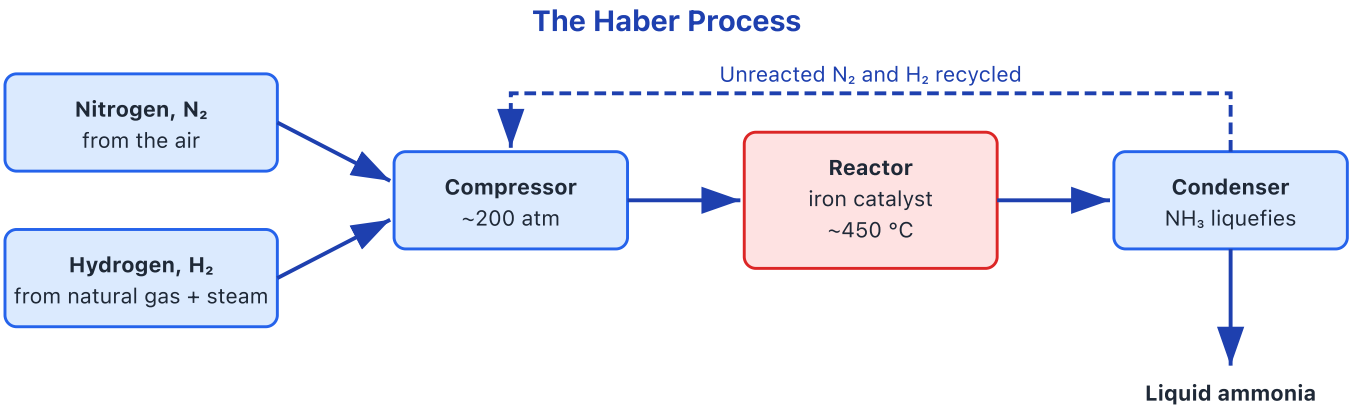
	Bacterial extraction (bioleaching)	Phytoextraction (phytomining)
How it works	Bacteria feed on low-grade ore (e.g. copper sulfide) and produce a solution of copper ions called a **leachate**. Copper is then obtained by displacement with scrap iron or by electrolysis.	**Plants** are grown on soil containing metal compounds; they absorb metal ions through their roots. The plants are harvested and burned; the **ash** contains metal compounds, from which the metal is extracted (e.g. by electrolysis or displacement).
Advantages	Uses low-grade ore and waste; less energy than smelting; no need to dig large open pits; no SO_2 from roasting sulfide ores.	Uses low-grade ore and contaminated land (cleans the soil); low energy; little landscape damage; burning plants releases energy that can be used.
Disadvantages	Very **slow**; the leachate can be toxic and must not escape into groundwater.	Very **slow** (growing seasons); needs large areas of land; yield of metal is small; depends on weather.

Exam Tip: "Evaluate" means give advantages *and* disadvantages and reach a judgement — e.g. biological methods are worthwhile for low-grade ores because they conserve high-grade ores and use less energy, even though they are slow.

The Haber Process — Rate vs Equilibrium (HT) — C6.1d, C6.1e, C6.1f

$$\mathbf{N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)} \quad \text{forward reaction exothermic } (\Delta H = -92 \text{ kJ/mol})$$

Conditions: **about 450 °C, about 200 atmospheres, iron catalyst**

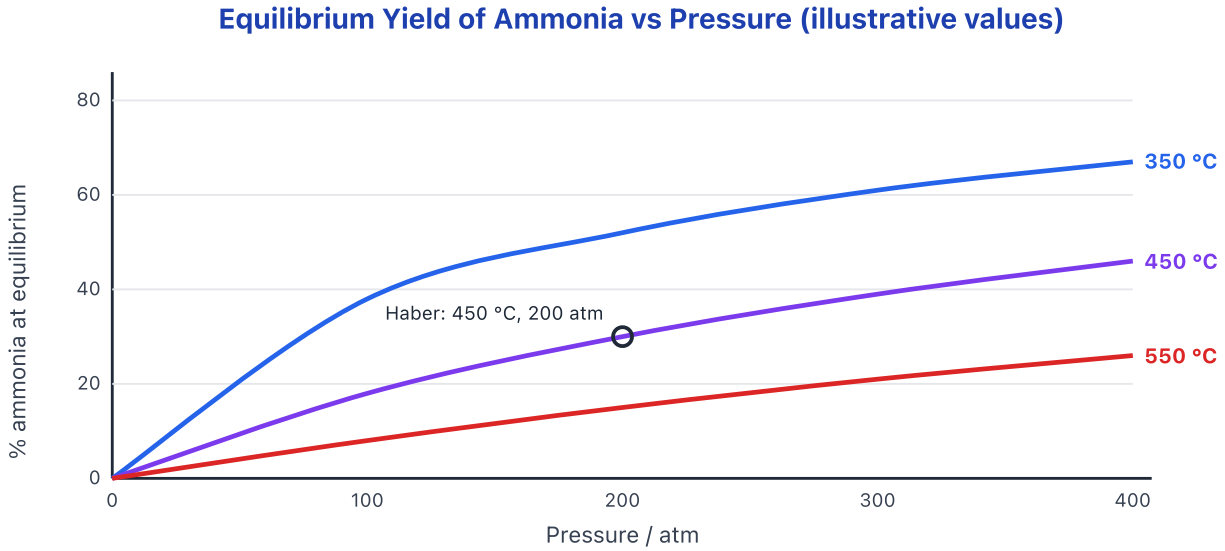


Only about 15% of the gases are converted on each pass, but recycling gives an overall conversion of about 97%

The trade-off (C6.1d): the conditions that give the *highest yield* are not the ones that give the *fastest rate*, so industry uses **compromise conditions**.

Condition	Effect on equilibrium yield	Effect on rate	Compromise chosen
Temperature	Forward reaction is exothermic, so a **low** temperature gives a higher yield	Low temperature makes the reaction too **slow**	~450 °C: reasonable yield at a reasonable rate
Pressure	4 gas molecules → 2, so **high** pressure gives a higher yield	High pressure increases the rate too	~200 atm: very high pressures need expensive, thick-walled plant and lots of energy, and are more dangerous
Catalyst (iron)	No effect on yield	Increases rate — equilibrium reached faster at a lower temperature	Always used; saves energy

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Interpreting graphs (C6.1e): on graphs like this, read across at a fixed pressure to compare temperatures (lower temperature → higher yield), or along one curve to see the effect of pressure (higher pressure → higher yield). On a graph of *rate* against temperature, rate rises steeply with temperature. Combining both explains the compromise.

The Contact Process (HT) — C6.1d, C6.1f

Sulfur burned in air: $S + O_2 \rightarrow SO_2$

$\mathbf{2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)}$ forward reaction exothermic

Conditions: **about 450 °C, about 1–2 atm, vanadium(V) oxide ($\mathbf{V_2O_5}$) catalyst**

SO_3 is absorbed in concentrated sulfuric acid (forming oleum, $H_2S_2O_7$), then diluted with water to give H_2SO_4

- Temperature: low favours SO_3 (exothermic) but is slow; ~450 °C is a compromise.
- Pressure: 3 gas molecules → 2, so high pressure would help, but the equilibrium yield is already **very high (over 95%) at atmospheric pressure**, so the extra cost of high pressure is not worth it.
- SO_3 is not added directly to water because the reaction is so exothermic it forms a corrosive mist.

Commercial conditions (C6.1f) depend on:

- **Availability and cost of raw materials** — N_2 from air is free, H_2 from natural gas costs money; recycling unreacted gases means no raw material is wasted.

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- **Energy supplies** — high temperatures and pressures need energy (cost and CO_2 emissions); the exothermic reaction's heat is used to preheat incoming gases.
- **Control of equilibrium position and rate** — chosen to give an acceptable yield quickly, as in the tables above; removing product (liquefying NH_3) pulls the equilibrium to the right.

Fertilisers — C6.1g, C6.1i

Plants need three main elements for healthy growth. Fertilisers replace the elements removed from the soil by crops, increasing **crop yield**.

Element	Why plants need it	Example compounds
Nitrogen (N)	To make proteins — healthy leaf and stem growth	Ammonium nitrate, ammonium sulfate, urea
Phosphorus (P)	Healthy root growth	Ammonium phosphate, superphosphate
Potassium (K)	Healthy flowers and fruit; resistance to disease	Potassium chloride, potassium sulfate

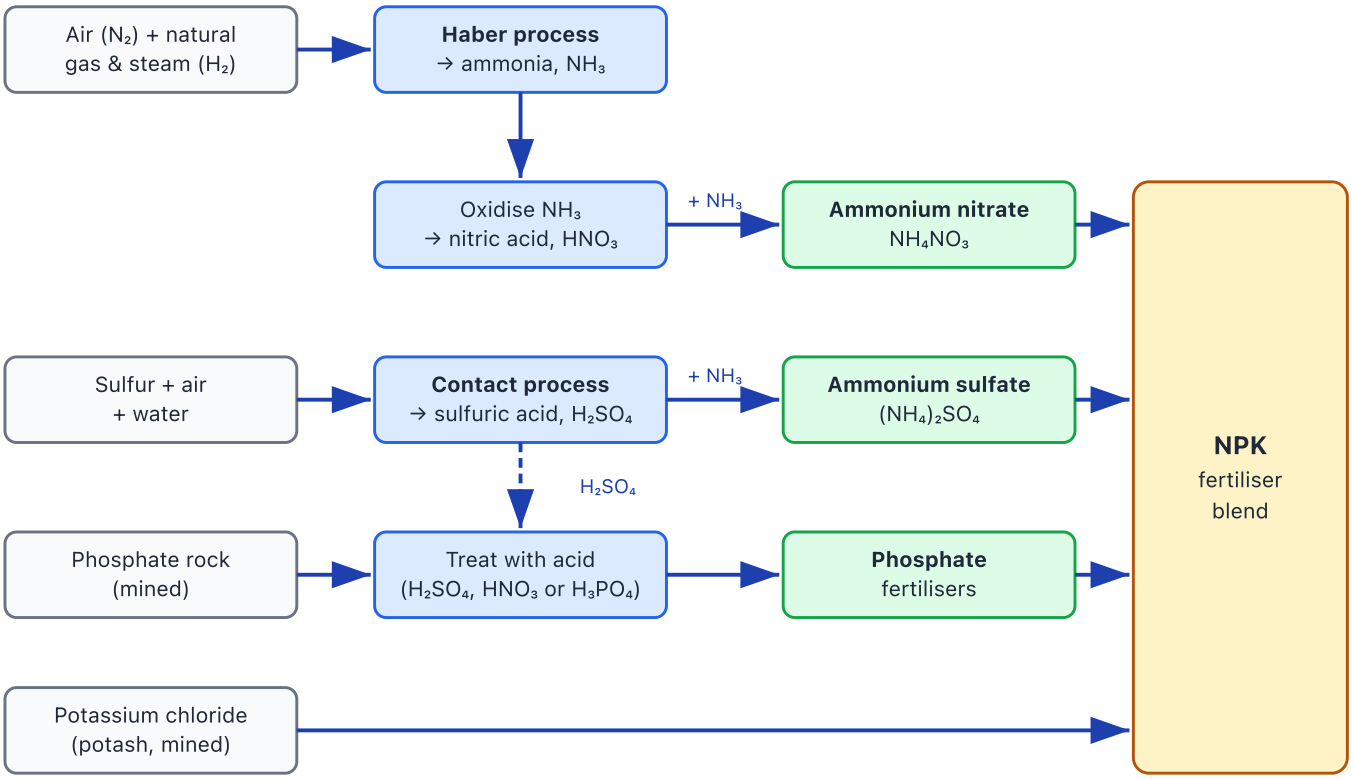
NPK fertilisers are formulations containing all three in chosen proportions.

Importance of the Haber process (C6.1g): plants cannot use nitrogen gas from the air directly. The Haber process "fixes" atmospheric nitrogen as ammonia, which is turned into nitrogen fertilisers. Without it, there would not be enough food for the world's population — around half the world's food is grown using fertilisers made from Haber-process ammonia.

Fertilisers: Integrated Industrial Production — C6.1j

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Making Fertilisers: Several Integrated Processes



$$NH_3 + HNO_3 \rightarrow NH_4NO_3 \text{ (ammonium nitrate)} \quad 2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4 \text{ (ammonium sulfate)}$$

Industrial vs Laboratory Production of Fertilisers — C6.1h

In the lab, **ammonium sulfate** (or potassium sulfate) is made by a titration method (PAG C6): titrate dilute ammonia solution (or potassium hydroxide) with dilute sulfuric acid using an indicator to find the volume needed; repeat *without* indicator using those volumes; then evaporate some water and leave to **crystallise**; filter and dry the crystals.

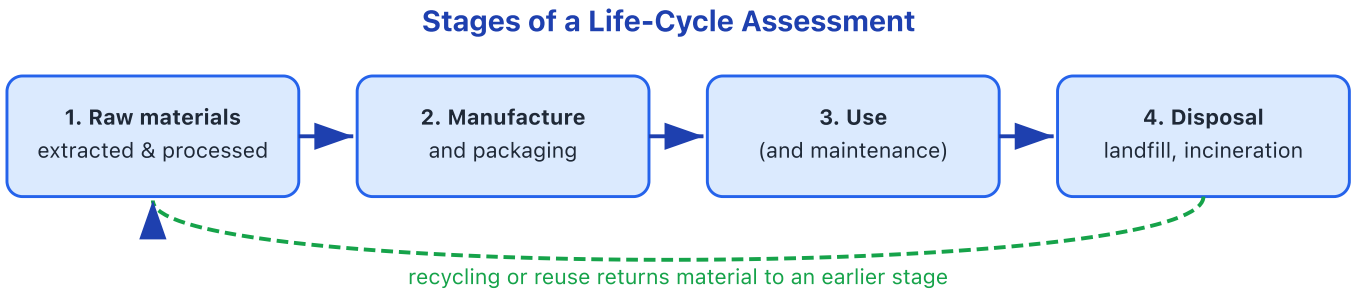
	Laboratory synthesis	Industrial production
Scale	Grams	Thousands of tonnes per day
Process type	Batch — one portion at a time	Continuous — runs day and night

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Raw materials	Bought ready-made: ammonia solution, dilute sulfuric acid	Made on site from basic raw materials (air, natural gas, sulfur, water) by several integrated processes
Stages	Few: titration, evaporation, crystallisation	Many: Haber process, Contact process, then neutralisation in large reactors
Energy	Bunsen / water bath	Heat from exothermic reactions is recovered and reused
Equipment / staff	Simple glassware, one person	Expensive plant, automated control, many workers

Life-Cycle Assessment — C6.1k, C6.1l

> A **life-cycle assessment (LCA)** assesses the environmental impact and use of resources (raw materials, energy, water) at **every stage** of a product's life.



Transport between every stage uses fuel and causes pollution — it is included in the assessment

Interpreting LCA data (C6.1l): compare the figures stage by stage, then overall. Remember that some impacts are easy to quantify (energy, CO_2, water) but others (e.g. visual pollution, effects of litter) need **value judgements**, so LCAs can be biased or selectively used, e.g. in advertising.

Stage (illustrative data)	Paper bag	Plastic (poly(ethene)) bag
Raw materials	Timber (renewable)	Crude oil (finite)

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Energy to make one bag / kJ	1 700	600
CO_2 emitted making one bag / g	60	35
Uses before disposal	Usually 1	Can be reused many times
Disposal	Biodegradable; easily recycled	Not biodegradable; litter; can be recycled

Illustrative figures for practising interpretation — real values vary between studies.

Recycling — C6.1m, C6.1n

Some materials are recycled for a **different use** (C6.1m):

- **Glass** bottles are crushed and melted to make new glass, or crushed for use in road surfaces ("glasphalt") and as building aggregate — viable because glass melts at a lower temperature than making it from sand, saving energy.
- **PET plastic bottles** are shredded, melted and spun into polyester fibres for fleece clothing and carpets — viable because PET is a thermoplastic that can be remelted, and it reduces landfill and crude oil use.
- **Scrap steel** is added to the furnace when making new steel; **aluminium** cans are melted and recast — recycling aluminium uses about 5% of the energy needed to extract it by electrolysis.

Factors affecting decisions on recycling (C6.1n):

- Energy needed to collect, sort and reprocess compared with extracting new material.
- Cost of collection, transport and separation (mixed materials are hard to separate).
- Value of the material and scarcity/finite nature of the raw material.
- Environmental impact: reduced mining and quarrying, less landfill, fewer emissions.
- Whether the quality of the recycled material is good enough for the new use.

Alloys — C6.1o

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An **alloy** is a mixture of a metal with one or more other elements (usually metals). The different-sized atoms distort the regular layers, so layers cannot slide as easily — alloys are usually **harder and stronger** than the pure metal.

Alloy	Composition	Properties	Uses
Steel (low-carbon / mild)	Iron + about 0.1–0.3% carbon	Strong, easily shaped	Car bodies, girders, wire
Steel (high-carbon)	Iron + about 0.6–1.5% carbon	Very hard but more brittle	Cutting tools, drill bits
Stainless steel	Iron + chromium (+ nickel)	Hard, resists corrosion	Cutlery, sinks, surgical instruments
Brass	Copper + zinc	Harder than copper, gold-coloured, does not corrode easily	Musical instruments, door fittings, plug pins
Bronze	Copper + tin	Hard, tough, resists corrosion	Statues, medals, ship propellers, bells
Solder	Tin + lead (traditional) or tin + copper/silver (lead-free)	Low melting point	Joining electrical components and pipes
Duralumin	Aluminium + about 4% copper (+ small amounts of magnesium, manganese)	Low density but much stronger than aluminium	Aircraft bodies, bicycle frames

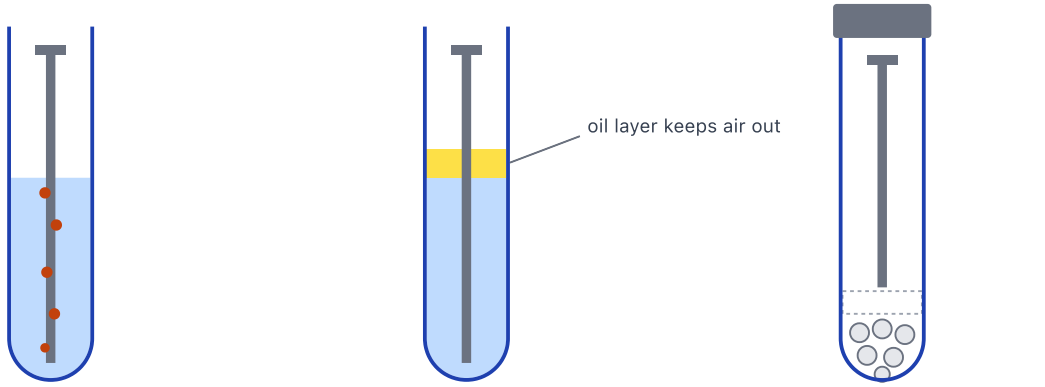
Corrosion — C6.1p

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Corrosion is the destruction of a metal by chemical reaction with substances in its environment. It is an **oxidation**. The corrosion of iron and steel is called **rusting**; it needs both **water and oxygen**.

iron + oxygen + water → hydrated iron(III) oxide (rust) $4Fe + 3O_2 + 2xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O$

What Does Iron Need to Rust?



A: air + water
Rusts
tap water, open to air

B: water, no air
No rust
boiled water + oil layer

C: air, no water
No rust
bung + anhydrous calcium chloride

- Rusting is faster in **salt water** (e.g. near the sea or on gritted roads) and at higher temperatures.
- **Other metals** corrode too, but not all are damaged: **aluminium** forms a thin, tough layer of aluminium oxide that sticks to the surface and protects the metal underneath. Copper slowly forms a green coating (patina). Rust is *flaky* and falls off, exposing fresh iron, so iron keeps corroding.

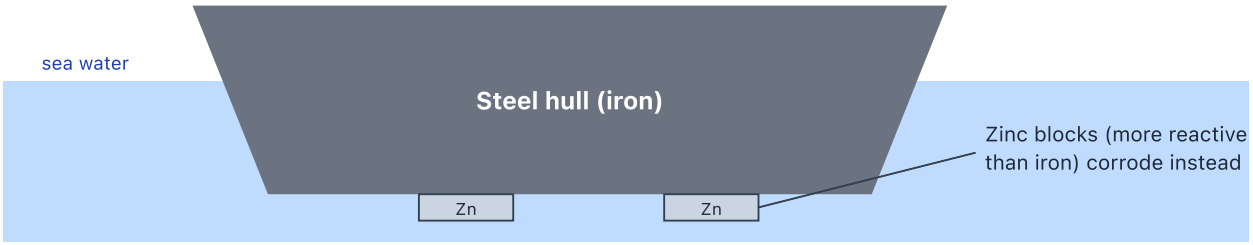
Preventing Corrosion — C6.1q

Method	How it works	Examples
Physical barrier	Coating keeps **oxygen and water** away from the metal surface	Paint (cars, bridges), oil/grease (moving parts), plastic coating (garden furniture, wire), electroplating with tin or chromium

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Sacrificial protection	A **more reactive** metal (zinc or magnesium) is attached; it is oxidised instead of the iron, so it corrodes ("is sacrificed") first	Zinc or magnesium blocks on ship hulls, pipelines, oil rigs
Galvanising	Coating of zinc — a barrier *and* sacrificial protection; still works if scratched	Buckets, car bodies, fences, roofing

Sacrificial Protection of a Steel Ship



$Zn \rightarrow Zn^{2+} + 2e^-$ Zinc is oxidised in preference to iron; the blocks are replaced when used up

Exam Tip: Tin plating (food cans) is only a **barrier**. Tin is *less* reactive than iron, so if the coating is scratched the iron rusts even faster. Zinc (galvanising) still protects when scratched because zinc is *more* reactive than iron.

Comparing Materials — C6.1r, C6.1s

Material (typical values)	Density / g/cm³	Tensile strength / MPa	Melting / softening point / °C	Electrical conductivity	Other properties
Glass ceramic (soda-lime glass)	2.5	~50	softens ~700	Insulator	Transparent, hard, brittle, unreactive

Clay ceramic (porcelain, brick)	2.0–2.4	~40	>1500	Insulator	Hard, brittle, strong in compression, heat resistant
Polymer: poly(ethene) (LDPE)	0.92	~10	~110	Insulator	Flexible, low density, unreactive, cheap
Polymer: poly(ethene) (HDPE)	0.96	~30	~130	Insulator	Stiffer and stronger than LDPE
Composite: carbon-fibre reinforced polymer	1.6	~600+	resin softens ~150–200	Conducts slightly	Very strong for its weight, expensive
Metal: steel (mild)	7.8	~400	~1450	Good conductor	Strong, malleable, ductile, rusts
Metal: aluminium	2.7	~90 (alloys up to ~500)	660	Very good conductor	Low density, does not corrode
Metal: copper	8.9	~220	1085	Excellent conductor	Ductile, does not corrode easily

Approximate typical values — real figures vary with the exact grade of material.

Choosing materials (C6.1s): match the property to the job and justify with data.

- **Aircraft body** — duralumin or carbon-fibre composite: high strength-to-weight ratio (low density, high tensile strength).
- **Electrical wiring** — copper (excellent conductor, ductile) with a PVC (polymer) insulating coat.

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- **Oven dishes, spark-plug insulators** — ceramics: very high melting point, insulators.
- **Windows** — glass: transparent, hard, scratch resistant (but brittle).
- **Bridges and buildings** — steel (strong, cheap) or reinforced concrete (a composite).
- **Carrier bags / packaging** — poly(ethene): flexible, low density, waterproof, cheap.

C6.2 Organic Chemistry

Functional Groups and Homologous Series — C6.2a

> A **homologous series** is a family of organic compounds with the **same functional group** and the **same general formula**. Successive members differ by **CH_2**; they have **similar chemical reactions** and a **gradual trend** in physical properties (e.g. boiling point increases with chain length). A **functional group** is the atom or group of atoms that determines how the molecule reacts.

Homologous series	Functional group	General formula	Name ending	Example
Alkanes	C–C single bonds only (saturated)	C_nH_{2n+2}	-ane	Ethane, C_2H_6
Alkenes	C=C double bond (unsaturated)	C_nH_{2n}	-ene	Ethene, C_2H_4
Alcohols	–OH (hydroxyl)	$C_nH_{2n+1}OH$	-ol	Ethanol, C_2H_5OH
Carboxylic acids	–COOH (carboxyl)	$C_nH_{2n+1}COOH$	-oic acid	Ethanoic acid, CH_3COOH

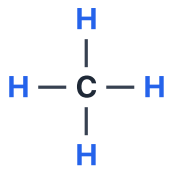
Name stems give the number of carbon atoms: **meth-** 1, **eth-** 2, **prop-** 3, **but-** 4, **pent-** 5. (For carboxylic acids the carbon in –COOH is counted.)

Displayed Formulae — C6.2b

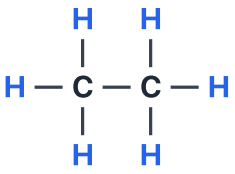
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A **displayed formula** shows every atom and every bond. Carbon always forms 4 bonds, hydrogen 1, oxygen 2.

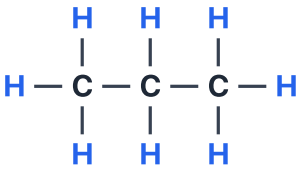
Displayed Formulae: The First Four Alkanes (C_nH_{2n+2})



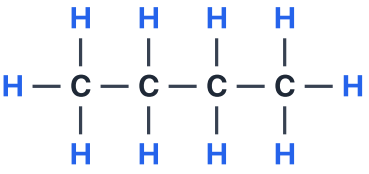
Methane · CH_4



Ethane · C_2H_6

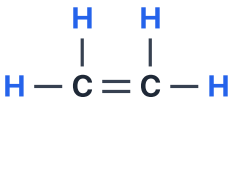


Propane · C_3H_8

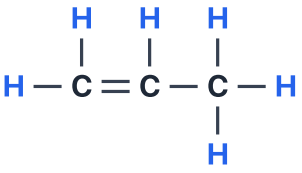


Butane · C_4H_{10}

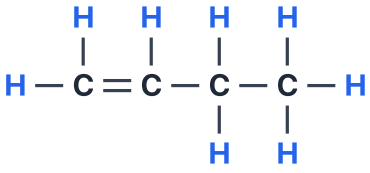
Displayed Formulae: The First Four Alkenes (C_nH_{2n})



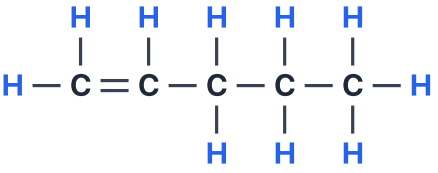
Ethene · C_2H_4



Propene · C_3H_6



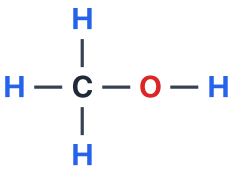
Butene (but-1-ene) · C_4H_8



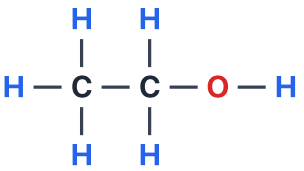
Pentene (pent-1-ene) · C_5H_{10}

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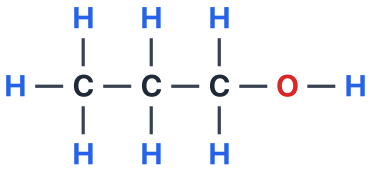
Displayed Formulae: The First Four Alcohols ($C_nH_{2n+1}OH$)



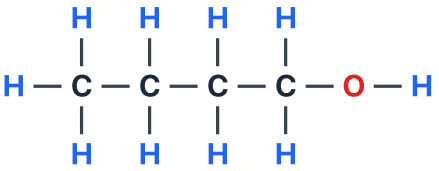
Methanol · CH_3OH



Ethanol · C_2H_5OH

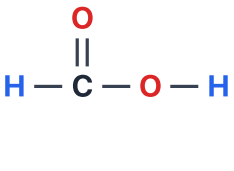


Propanol (propan-1-ol) · C_3H_7OH

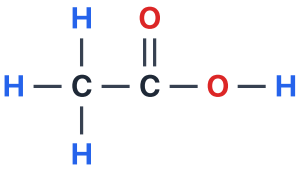


Butanol (butan-1-ol) · C_4H_9OH

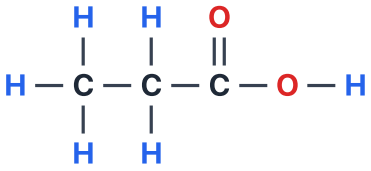
Displayed Formulae: The First Four Carboxylic Acids (functional group –COOH)



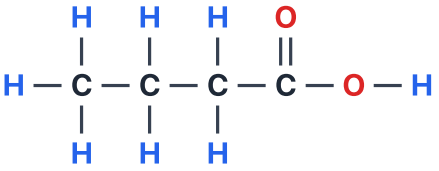
Methanoic acid · HCOOH



Ethanoic acid · CH_3COOH



Propanoic acid · C_2H_5COOH



Butanoic acid · C_3H_7COOH

> **Exam Tip:** Count bonds on every carbon — it must have exactly four (a double bond counts as two). In a carboxylic acid the –COOH carbon has a C=O *and* a C–O–H. Methanoic acid has only one carbon.

Reactions of Alkanes, Alkenes and Alcohols — C6.2c

Combustion. All three series burn. Complete combustion (plenty of oxygen) gives **carbon dioxide and water**; incomplete combustion gives carbon monoxide and/or soot (carbon). Alkenes burn with a smokier flame than alkanes.

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$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O \quad \cdot \quad 2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$$
$$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O \quad \cdot \quad C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$$

Addition reactions of alkenes. The C=C double bond opens and atoms add across it, making a saturated product.

Addition of Bromine Across the C=C Double Bond



Ethene (colourless) **Bromine** (orange) **1,2-dibromoethane** (colourless)

With hydrogen (nickel catalyst, ~150 °C) the same addition gives ethane, C_2H_6

- **Bromine:** $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$ (1,2-dibromoethane). Bromine water turns from **orange to colourless** — the test for an alkene (C=C). Alkanes do not decolourise it.
- **Hydrogen:** $C_2H_4 + H_2 \rightarrow C_2H_6$ (nickel catalyst, about 150 °C) — the alkene becomes an alkane. Propene + hydrogen → propane.
- Other members react the same way: propene + bromine → 1,2-dibromopropane, $C_3H_6Br_2$.

Oxidation of alcohols. Warming an alcohol with acidified **potassium manganate(VII)** oxidises it to the **carboxylic acid** with the same number of carbon atoms. The purple solution turns colourless.

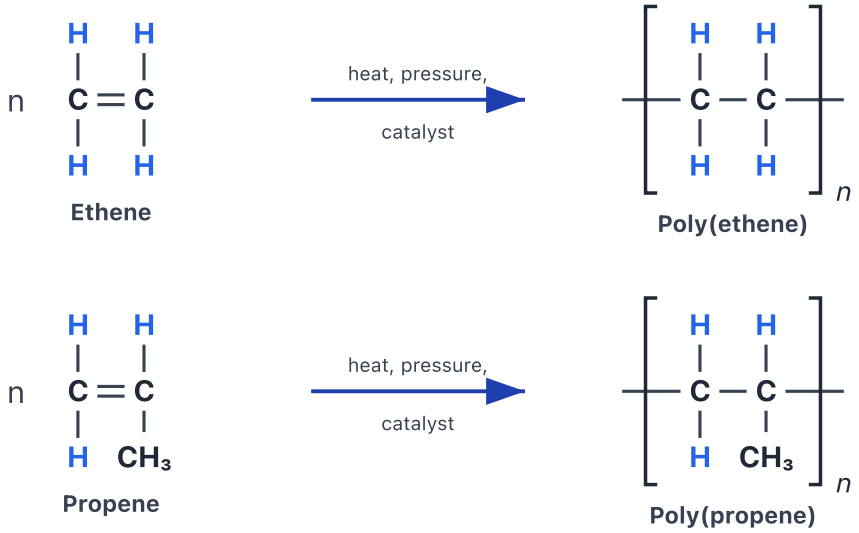
$$\text{ethanol} \rightarrow \text{ethanoic acid} \quad CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O \quad ([O] = \text{oxygen from the oxidising agent})$$
$$\text{propanol} \rightarrow \text{propanoic acid} \quad \cdot \quad \text{butanol} \rightarrow \text{butanoic acid}$$

Addition Polymerisation — C6.2d, C6.2g

> In **addition polymerisation**, many small **alkene monomers** join together. The **C=C double bond** in each monomer opens and the monomers link into a long saturated chain. The

polymer is the **only product**.

Addition Polymerisation: Monomer → Repeat Unit



- **Monomer → polymer:** draw the two C=C carbons with a single bond between them, keep their other atoms/groups in place, add bonds sticking out through brackets on each side and write *n*.
- **Polymer → monomer:** take the repeat unit, remove the extension bonds and brackets, and put a double bond between the two carbons.
- Other examples: chloroethene → poly(chloroethene) (PVC); tetrafluoroethene → poly(tetrafluoroethene) (PTFE).

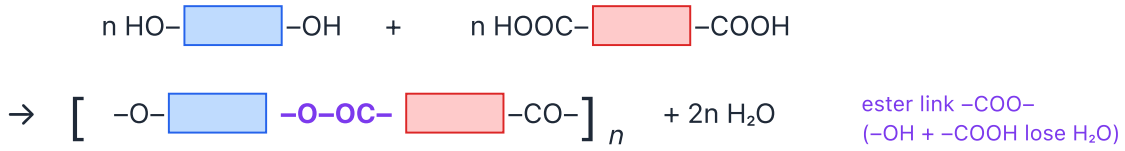
Condensation Polymerisation (HT) — C6.2e

In **condensation polymerisation**, monomers with **two functional groups each** (the minimum needed so the chain can keep growing at both ends) react together, and a **small molecule** such as water (or HCl) is given off every time a link forms. Usually there are two different monomers, each with two of the same functional group.

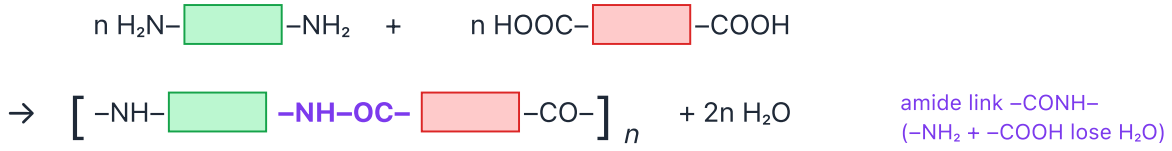
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Condensation Polymers Shown as Block Diagrams

Polyester (e.g. PET): diol + dicarboxylic acid



Polyamide (e.g. nylon): diamine + dicarboxylic acid



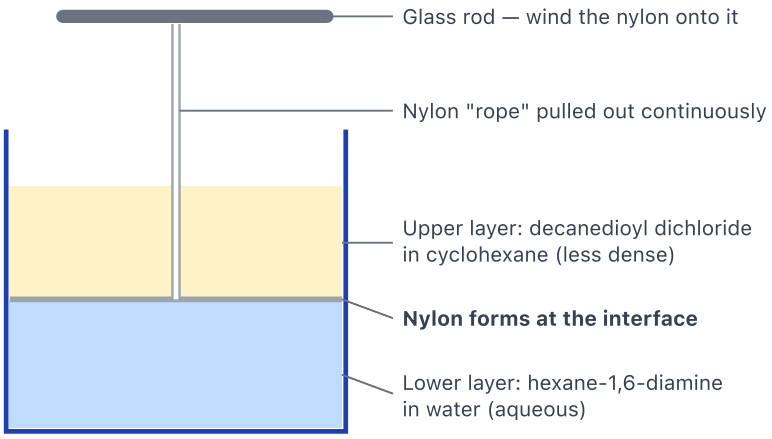
Each block stands for the carbon chain of a monomer; only the functional groups are drawn

	Addition polymerisation	Condensation polymerisation
Monomers	One type, containing C=C	Usually two types, each with two functional groups
Products	Polymer only	Polymer + small molecule (e.g. H_2O) for every link formed
Repeat unit	Same atoms as the monomer	Contains one of each monomer minus the small molecule(s)

Exam Tip: For n molecules of each monomer, about **2n** small molecules are lost, because each repeat unit contains **two** new links. Make sure the ends of each block keep the right number of bonds.

Making Nylon in the Lab (HT) — C6.2f

Making Nylon: the "Nylon Rope Trick"



1. Pour a solution of hexane-1,6-diamine (in water) into a small beaker.
2. Carefully pour a solution of decanedioyl dichloride (in cyclohexane) down the side, so it floats on top without mixing.
3. Nylon forms where the two layers meet. Grip it with tweezers and pull it out slowly, winding it onto a glass rod; the polymer keeps forming as fresh solutions meet.
4. Wash the nylon with water before handling. The small molecule given off here is HCl. Wear gloves and eye protection; the reagents are corrosive/harmful and cyclohexane is flammable — use a fume cupboard.

Naturally Occurring Polymers — C6.2h

Polymer	Monomers
DNA (deoxyribonucleic acid)	Four different **nucleotides**, named after their bases: **adenine (A), thymine (T), cytosine (C), guanine (G)**. Two polymer strands form a double helix.
Proteins	**Amino acids** (joined by condensation polymerisation — proteins are polyamides)
Starch, cellulose	**Sugars** (glucose)

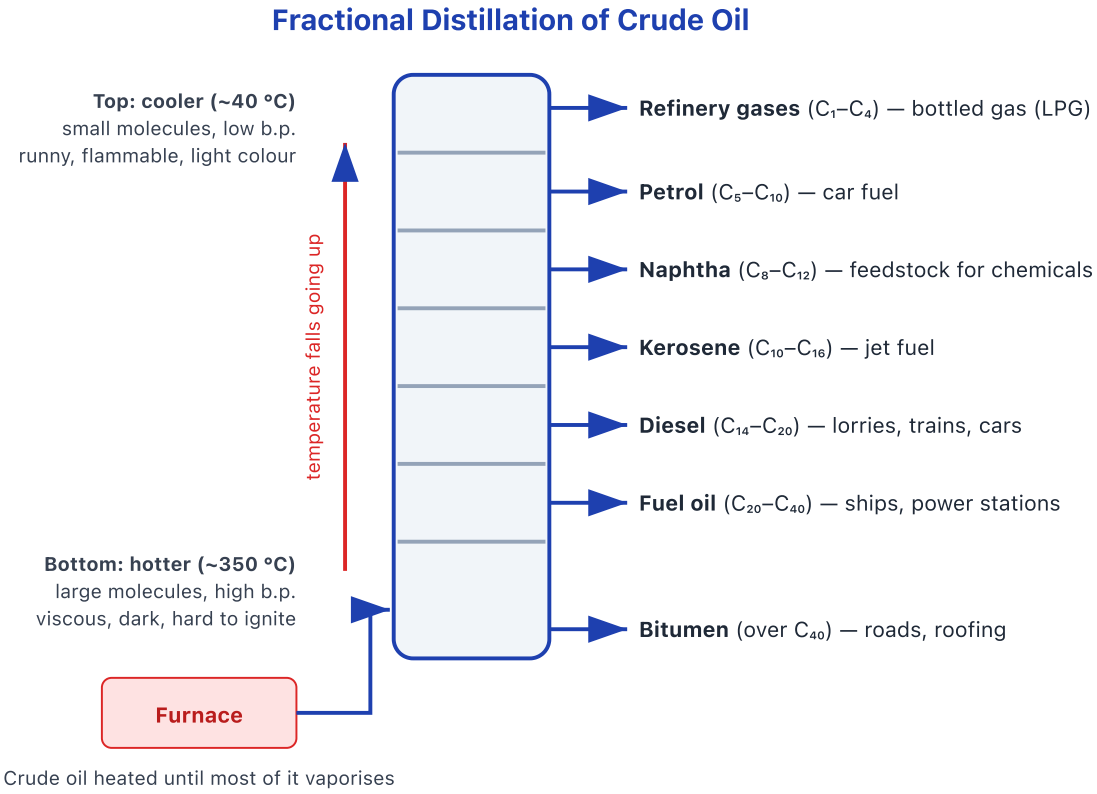
Generality of Reactions — C6.2i

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The reactions of an organic compound are determined by its **functional group**, not by the length of its carbon chain. So if you know how ethene reacts with bromine, you can predict that any alkene will react the same way; if ethanol is oxidised to ethanoic acid, propanol is oxidised to propanoic acid. This is why compounds are grouped into homologous series.

Crude Oil and Fractional Distillation — C6.2j, C6.2k, C6.2l

> **Crude oil** is a mixture of many hydrocarbons, mostly **alkanes** (C_nH_{2n+2}). It is separated into **fractions** — groups of hydrocarbons with similar numbers of carbon atoms and similar boiling points — by **fractional distillation**.



Describe (C6.2j): crude oil is heated until it vaporises and the vapour enters the bottom of the column. The column is **hot at the bottom and cooler at the top**. Vapours rise; each fraction **condenses** at the level where the temperature is just below its boiling point and is piped off. Very large molecules (bitumen) do not vaporise and are run off at the bottom; the smallest stay as gases and leave at the top.

Explain (C6.2k): larger alkane molecules have **stronger intermolecular forces** between them (more contact area), so more energy is needed to separate them — they have **higher boiling**

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points and condense lower down. Smaller molecules have weaker intermolecular forces, lower boiling points, and rise higher. The *covalent bonds* inside the molecules are not broken.

Crude Oil as a Resource — C6.2m, C6.2n

- Crude oil is the main source of **hydrocarbons** and the **feedstock** (raw material) for the **petrochemical industry**, which makes plastics, solvents, lubricants, detergents, dyes and medicines.
- Modern life depends on hydrocarbons: fuels for transport, heating and electricity generation, and the materials listed above.
- Crude oil is a **finite (non-renewable)** resource: it formed over millions of years from the remains of ancient marine organisms and is being used far faster than it forms. Burning it also releases CO_2. This is why alternatives (biofuels, electric vehicles, recycling of plastics) matter.

Cracking — C6.2o

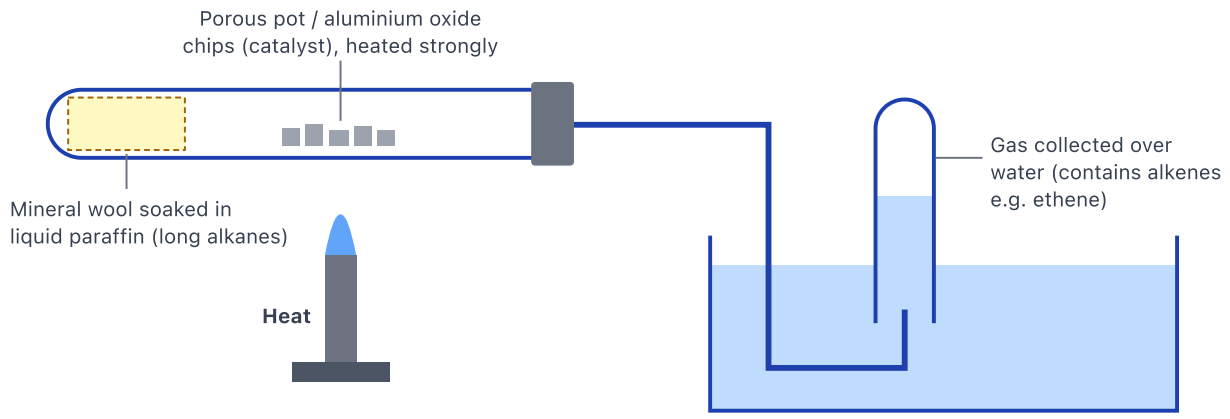
> **Cracking** is the **thermal decomposition** of long-chain alkanes into **shorter-chain alkanes** (useful as fuels such as petrol) and **alkenes** (used to make polymers and other chemicals).

$$C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4 \quad \text{(decane} \rightarrow \text{octane + ethene)} \quad \cdot \quad C_{16}H_{34} \rightarrow C_8H_{18} + C_4H_8 + 2C_2H_4$$

- **Conditions:** the long-chain hydrocarbons are heated to vaporise them, then passed over a **hot catalyst** (zeolite, or silica/aluminium oxide) at about 500 °C — *catalytic cracking*; or mixed with steam and heated to a very high temperature (over 800 °C) — *steam cracking*.
- **Reasons:** fractional distillation gives **more long-chain hydrocarbons than are needed** and **not enough short-chain ones** (petrol is in high demand). Cracking matches supply to demand, and produces **alkenes**, which are reactive and used to make polymers (e.g. poly(ethene)) and ethanol.

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Cracking Liquid Paraffin in the Laboratory



Remove the delivery tube from the water before you stop heating, to prevent suck-back

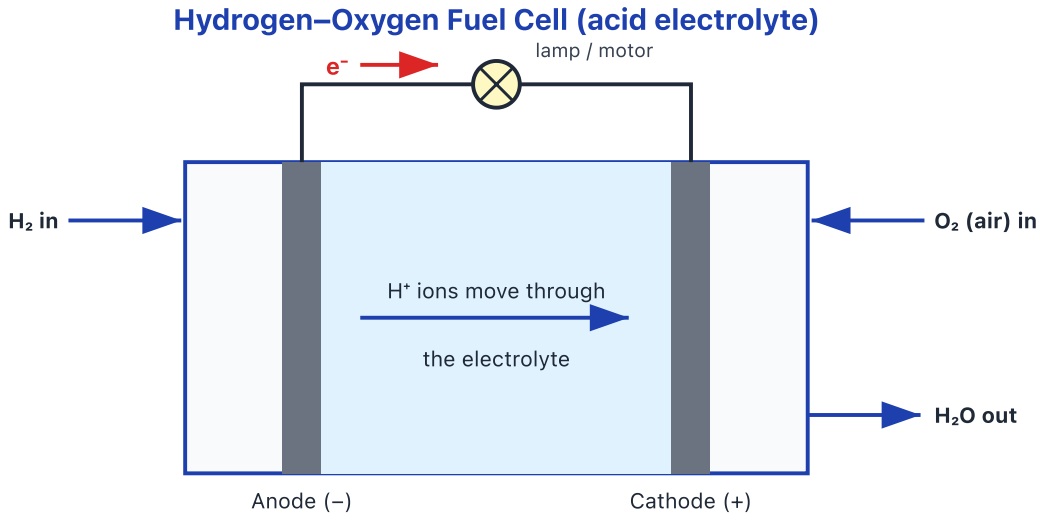
Exam Tip: To show the gas contains alkenes, shake it with **bromine water**: it turns from orange to colourless. Liquid paraffin itself (alkanes) does not decolourise bromine water.

Chemical Cells and Fuel Cells — C6.2p, C6.2q

A **chemical cell** (battery) produces a potential difference from a chemical reaction between its contents. It produces a voltage **until one of the reactants is used up**; then it is "flat". (Rechargeable cells can have the reaction reversed by an external current.)

A **fuel cell** is a cell supplied continuously with a fuel and oxygen (or air); the fuel is oxidised electrochemically to produce a potential difference. It keeps working **as long as fuel is supplied**.

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Anode: $H_2 \rightarrow 2H^+ + 2e^-$ · Cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$ · Overall: $2H_2 + O_2 \rightarrow 2H_2O$

Hydrogen–oxygen fuel cell: advantages	Disadvantages
Only product is **water** — no CO_2 or pollutants at the point of use	Hydrogen is a flammable gas, hard to **store** (high-pressure tanks) and transport
Does not run down or need recharging — refuel in minutes	Most hydrogen is made from fossil fuels (or electrolysis using electricity that may come from fossil fuels)
More efficient than burning fuel in an engine; few moving parts, quiet	Fuel cells are expensive (platinum catalyst); few refuelling stations
No toxic metal waste at end of life (unlike many batteries)	Performance can drop at low temperatures

Other fuel cells and cells for given uses: e.g. methanol fuel cells are easier to refuel (liquid fuel) but produce CO_2. For a mobile phone, a *rechargeable lithium-ion battery* is more practical (small, no fuel supply needed). For a bus or a spacecraft that needs long range and quick refuelling, a hydrogen fuel cell may be better — in spacecraft the water produced is drunk by astronauts.

C6.3 Interpreting and Interacting with Earth Systems

The Early Atmosphere — C6.3a

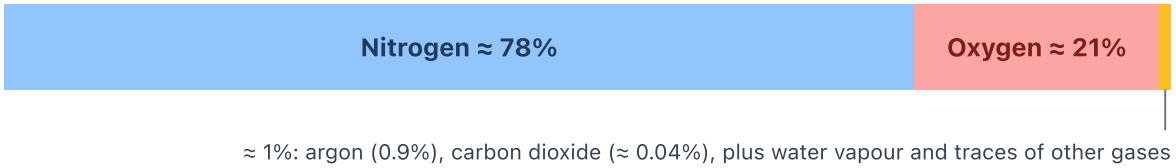
Evidence about the atmosphere billions of years ago is limited (there were no observers; evidence comes from rocks, fossils and comparison with other planets and volcanoes today), so the theories are uncertain. One accepted model:

- In the first billion years, intense **volcanic activity** released gases that formed the early atmosphere: mainly **carbon dioxide**, with **water vapour**, nitrogen and small amounts of methane and ammonia. There was **little or no oxygen** — similar to the atmospheres of Mars and Venus today.
- As the Earth cooled, water vapour **condensed to form the oceans**.
- Carbon dioxide **dissolved in the oceans**; carbonates precipitated forming sediments, and later marine organisms locked carbon into shells and skeletons (limestone) and fossil fuels. So the amount of CO_2 in the atmosphere decreased.

Development of an Oxygen-Rich Atmosphere — C6.3b

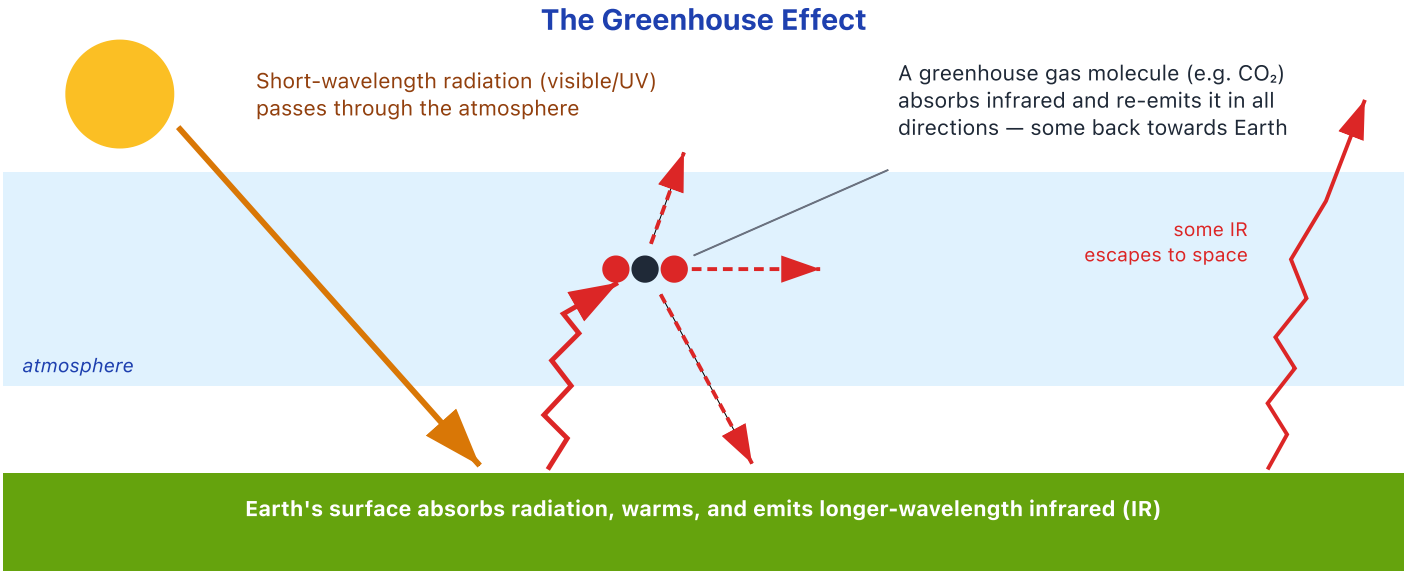
- About 2.7 billion years ago, **algae (and later plants)** evolved and produced oxygen by **photosynthesis**: **$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$**. This also removed more CO_2.
- Oxygen levels rose gradually over hundreds of millions of years, allowing animals to evolve. An ozone layer formed, protecting land life from UV.
- Nitrogen built up (released by volcanoes, from ammonia, and by denitrifying bacteria) and, being very unreactive, stayed in the atmosphere.

Composition of Dry Air Today (by volume)



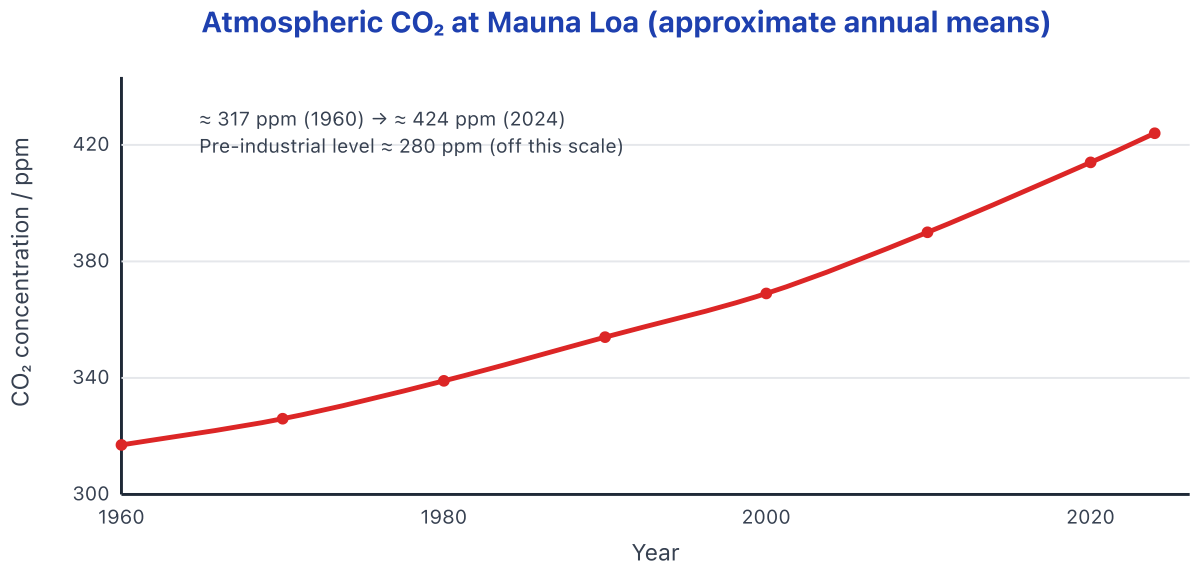
≈ 1%: argon (0.9%), carbon dioxide (≈ 0.04%), plus water vapour and traces of other gases

The Greenhouse Effect — C6.3c



- The Sun emits mostly **short-wavelength** radiation, which passes through the atmosphere and is absorbed by the Earth's surface, warming it.
- The warm Earth emits **longer-wavelength infrared** radiation. **Greenhouse gases** — carbon dioxide, methane and water vapour — **absorb** this infrared (their bonds vibrate) and then **re-emit** it in all directions, including back towards the surface, which warms the lower atmosphere.
- The natural greenhouse effect keeps the Earth's mean temperature about 33 °C warmer than it would otherwise be — essential for life. Increasing greenhouse gas concentrations **enhance** it.

Evidence for Human-Caused Climate Change — C6.3d



- **Evidence:** atmospheric CO_2 has risen from about 280 ppm before the Industrial Revolution to over 420 ppm today. The rise **correlates** with increasing consumption of fossil fuels (and deforestation). Global mean temperature has risen by over 1 °C over the same period. Ice cores show past CO_2 levels and temperatures changing together.
- **Uncertainties:** a correlation does not on its own prove cause; past data (from ice cores, tree rings) are indirect and less precise; there are natural causes of variation (volcanic eruptions, changes in the Sun's output, ocean cycles); the climate system is very complex, so computer models make assumptions; data were collected with different methods over time and location.
- Despite uncertainties, the great majority of climate scientists, using peer-reviewed evidence, agree that human activities are the main cause of recent warming.

> **Exam Tip:** CM6.3ii: use orders of magnitude. CO_2 is only ~0.04% of the air, but a rise from 280 to 420 ppm is a **50% increase** — small percentages of a strong absorber still matter.

Effects of Increased CO_2 and Methane, and Mitigation — C6.3e

Gas	Main human sources
Carbon dioxide	Burning fossil fuels (power stations, vehicles, industry), deforestation, cement making
Methane	Cattle (digestion), rice paddies, decay of waste in landfill, leaks from gas extraction

Potential effects: rising global temperatures; melting glaciers and polar ice and **sea-level rise** (also from thermal expansion of water) causing flooding of low-lying areas; more frequent and severe **extreme weather** (storms, droughts, heatwaves); changes in rainfall affecting food production and water supplies; changes to the distribution and survival of species (habitats); **ocean acidification** as more CO_2 dissolves.

Mitigation	Scale, risk and environmental implications
Use renewable energy / nuclear instead of fossil fuels	Large investment; renewables are intermittent; nuclear waste and public concern

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Improve energy efficiency, insulate homes, use public transport	Needs changes by millions of people; relatively low risk
Carbon capture and storage (CO_2 stored underground)	Expensive, energy-intensive, risk of leaks; technology still being scaled up
Plant trees (reforestation); stop deforestation	Needs large areas of land; slow; trees can burn or be cut down
Reduce methane (capture landfill gas, change diets/farming)	Methane is a stronger greenhouse gas, so cuts have a big effect; social/economic resistance
Carbon taxes, emission caps, international agreements	Requires global cooperation; can be politically difficult and costly for some countries

Atmospheric Pollutants — C6.3f

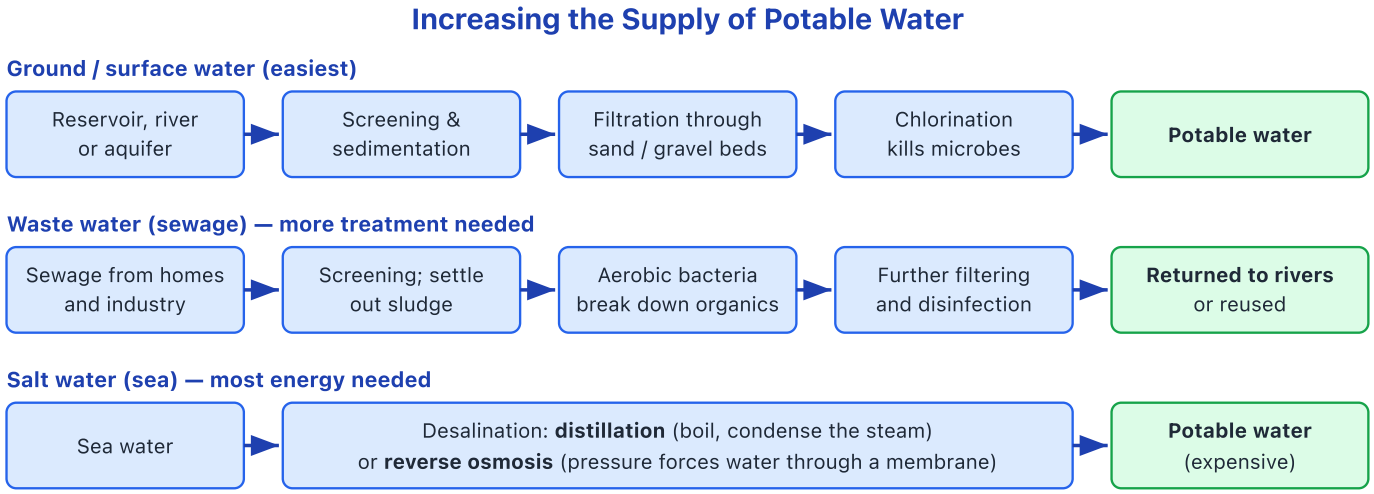
Pollutant	Major sources	Problems caused	Reduction
Carbon monoxide, CO	Incomplete combustion of carbon-containing fuels (vehicles, faulty boilers)	**Toxic**: colourless, odourless; combines with haemoglobin so blood carries less oxygen — can kill	Catalytic converters ($CO \rightarrow CO_2$); servicing boilers; CO alarms
Sulfur dioxide, SO_2	Burning fuels containing sulfur impurities (coal, some oil) in power stations, ships	**Acid rain** (damages buildings, kills trees and aquatic life); respiratory problems	Remove sulfur from fuels; flue gas desulfurisation (react with calcium carbonate/oxide)
Oxides of nitrogen, NO_x	Nitrogen and oxygen from the air react at the **high temperatures** in engines and furnaces	**Acid rain**; photochemical smog; breathing problems, aggravates asthma	Catalytic converters ($NO_x \rightarrow N_2$)

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Particulates (soot, carbon; also unburnt hydrocarbons)	Incomplete combustion, especially **diesel** engines; burning wood/coal	Lung and heart disease, cancer; **global dimming** (reflect sunlight); blacken buildings	Particulate filters on diesel vehicles; cleaner fuels

Potable Water — C6.3g

> **Potable water** is water that is **safe to drink**: it has low levels of dissolved salts and microbes. It is *not* chemically pure — it still contains some dissolved substances.



- **Ground and surface water** (rain collected in reservoirs, rivers, aquifers) is the easiest to treat: **sedimentation** and **filtration** remove solids; **chlorination** kills microbes.
- **Waste water** contains organic matter and harmful microbes, so it needs more stages (screening, sedimentation, biological treatment) before it can be safely returned to the environment or reused.
- **Salt water** needs **desalination**, which uses a great deal of energy, so it is expensive — used mainly in dry countries with plentiful energy but little fresh water.

Practical Activity Groups Linked to C6

PAG	Title	Where it fits in C6
PAG C1	Reactivity trend	Extraction of copper by heating copper(II) oxide with carbon (C6.1a)
PAG C2	Electrolysis	Electrolysis of aqueous sodium chloride and aqueous copper(II) sulfate, testing the products (C6.1b)
PAG C6	Titration	Preparation of potassium sulfate or ammonium sulfate using a titration method (C6.1h)

Quick Summary — Key Equations & Spec Checklist for C6

Process / idea	Equation or key fact
Iron extraction	$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
Copper by carbon reduction	$2CuO + C \rightarrow 2Cu + CO_2$
Aluminium electrolysis	$2Al_2O_3 \rightarrow 4Al + 3O_2$ (in molten cryolite)
Haber process	$N_2 + 3H_2 \rightleftharpoons 2NH_3$ · 450 °C, 200 atm, iron catalyst
Contact process	$2SO_2 + O_2 \rightleftharpoons 2SO_3$ · 450 °C, 1–2 atm, V_2O_5 catalyst
Fertilisers	$NH_3 + HNO_3 \rightarrow NH_4NO_3$ · $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$
Rusting	Iron + oxygen + water → hydrated iron(III) oxide
Test for alkenes	Bromine water: orange → colourless
Cracking	$C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$
Fuel cell	$2H_2 + O_2 \rightarrow 2H_2O$
Photosynthesis	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

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Spec	You should be able to...
C6.1a	Explain extraction of metals using carbon, including a non-ferrous metal (PAG C1)
C6.1b	Explain why and how electrolysis is used to extract some metals (PAG C2)
C6.1c (HT)	Evaluate bacterial extraction and phytoextraction
C6.1d (HT)	Explain the rate vs equilibrium trade-off in the Haber and Contact processes
C6.1e (HT)	Interpret graphs of reaction conditions versus rate (and yield)
C6.1f (HT)	Relate commercial conditions to raw materials, energy, equilibrium position and rate
C6.1g	Explain the importance of the Haber process in agriculture
C6.1h	Compare industrial and laboratory production of fertilisers (PAG C6)
C6.1i	Recall the importance of N, P and K compounds
C6.1j	Describe fertiliser production as integrated processes (ammonium nitrate, ammonium sulfate)
C6.1k / C6.1l	Describe the stages of a life-cycle assessment and interpret LCA data
C6.1m / C6.1n	Describe recycling for a different use; evaluate factors affecting recycling decisions
C6.1o	Describe composition, properties and uses of steel, brass, bronze, solder, duralumin
C6.1p / C6.1q	Describe corrosion and its conditions; explain barrier and sacrificial protection
C6.1r / C6.1s	Compare properties of glass and clay ceramics, polymers, composites and metals quantitatively; select materials for uses

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C6.2a / C6.2b	Recognise functional groups and homologous series; name and draw the first four alkanes, alkenes, alcohols and carboxylic acids
C6.2c	Predict products of combustion, addition of Br_2 and H_2, and oxidation of alcohols with potassium manganate(VII)
C6.2d / C6.2g	Recall addition polymerisation; deduce polymer from monomer and vice versa
C6.2e (HT)	Explain condensation polymerisation using block diagrams (polyesters, polyamides)
C6.2f (HT)	Describe how to make a polymer by condensation (nylon)
C6.2h	Recall DNA (four nucleotides) and polymers based on sugars and amino acids
C6.2i	Recall that functional groups determine the reactions of organic compounds
C6.2j / C6.2k / C6.2l	Describe and explain fractional distillation; fractions are mainly alkanes C_nH_{2n+2}
C6.2m / C6.2n	Crude oil as feedstock; modern dependence on hydrocarbons; finite resource
C6.2o	Describe cracking: conditions, reasons and useful products
C6.2p / C6.2q	Chemical cells run until reactants are used up; evaluate hydrogen–oxygen and other fuel cells
C6.3a / C6.3b	Interpret evidence for the early atmosphere; describe how oxygen built up
C6.3c	Describe the greenhouse effect in terms of radiation and matter
C6.3d / C6.3e	Evaluate evidence for human-caused climate change and its uncertainties; effects of CO_2 and CH_4 and mitigation
C6.3f	Sources and problems of CO, SO_2, NO_x and particulates
C6.3g	Methods of increasing potable water from ground, waste and salt water

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